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1. a) Carry out the following matrix-vector multiplication.

$$\begin{pmatrix} -1 & 2 & 4 & 3 \\ 1 & 0 & 8 & -1 \\ 0 & 0 & 2 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 35 \\ 37 \\ 22 \end{pmatrix}$$

- b) Give an example of a matrix that represents a linear transformation from $\mathbb{C}^7$ to $\mathbb{C}^2$ (with the understanding that bases have been chosen).

eg.

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

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2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 4 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}$$

CD , CA , DA , $A+B$

$$CD = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 0 & 3 \end{pmatrix}$$

$CA =$ undefined

$$DA = \begin{pmatrix} 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 0 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}$$

$$A+B = \begin{pmatrix} -2 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

b) Give an example of two matrices A and B —neither of them an identity matrix—so that their product AB is the following matrix M ; or explain why this is not possible.

$$M = \begin{pmatrix} 4 & 6 \\ 6 & 8 \end{pmatrix}$$

eg.

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 6 & 8 \end{pmatrix}$$

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3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

project to the line $x = 0$ and then rotate by 23° counterclockwise

$$\begin{pmatrix} \cos 23^\circ & -\sin 23^\circ \\ \sin 23^\circ & \cos 23^\circ \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\sin 23^\circ \\ 0 & \cos 23^\circ \end{pmatrix}$$

- b) Find the 3×3 matrix for the following linear transformation using the standard basis for $\mathbb{R}^3$.

project to the xz -plane

$$\begin{matrix} & x & y & z \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

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4. a) Use row reduction to put the following real matrix in reduced row echelon form.

$$\begin{pmatrix} 0 & 8 & 8 & 8 \\ 0 & 0 & 0 & 10 \\ 4 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

b) Write the following linear system over $\mathbb{R}$ as an augmented matrix. Then use row reduction to find the general solution to the system and write it in vector form.

$$w + 2x + 4y - 2z = 0$$

$$w + 4x - 2y = 0$$

$$\left(\begin{array}{cccc|c} 1 & 2 & 4 & -2 & 0 \\ 1 & 4 & -2 & 0 & 0 \end{array} \right) \xrightarrow{RR} \left(\begin{array}{cccc|c} 1 & 0 & 10 & -4 & 0 \\ 0 & 1 & -3 & 1 & 0 \end{array} \right)$$

$$\leadsto \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} -10 \\ 3 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 4 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

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5. a) Compute the inverse of the following matrix.

$$\begin{pmatrix} 4 & 0 & 2 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 0 & 2 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -1/4 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$$

b) Give an example of a non-identity matrix A where its inverse is equal to its square, A^2 ; or explain why this is not possible.

eg. $A = \begin{pmatrix} \cos 2\pi/3 & -\sin 2\pi/3 \\ \sin 2\pi/3 & \cos 2\pi/3 \end{pmatrix}$

since $A^2 = \begin{pmatrix} \cos(2 \cdot 2\pi/3) & -\sin(2 \cdot 2\pi/3) \\ \sin(2 \cdot 2\pi/3) & \cos(2 \cdot 2\pi/3) \end{pmatrix}$

and $A^3 = \underline{I} \Leftrightarrow A^2 = A^{-1}$

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6. a) Give a basis for the column space of A .

$$A = \begin{pmatrix} 1 & 4 & 3 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix} \xrightarrow{RR} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

2 pivot columns

So a basis for

col A is $\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} \right\}$

b) Give an example of a basis for the subspace of $\mathcal{P}_2$ consisting of the polynomials that have 8 as a root. (Recall that $\mathcal{P}_2$ is the vector space of real polynomials of degree at most 2.)

Every element of $\mathcal{P}_2$ has the form $p(x) = ax^2 + bx + c$,
using the basis $\{1, x, x^2\}$ for $\mathcal{P}_2$.

If $p(8) = 0$, we have $64a + 8b + c = 0$.

This is solved by $\begin{pmatrix} a \\ b \\ c \end{pmatrix} = b \begin{pmatrix} -\frac{1}{8} \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} -\frac{1}{64} \\ 0 \\ 1 \end{pmatrix}$

so $\left\{ \begin{pmatrix} -\frac{1}{8} \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{1}{64} \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for the subspace;

aka the polynomials

$-\frac{1}{8}x^2 + x$ and $-\frac{1}{64}x^2 + 1$
(or $x^2 - 8x$ and $x^2 - 64$) $\leftarrow$ re-scaled

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7. a) A linear operator on $\mathbb{R}^2$ is given by the following matrix in the standard basis:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

Choose a different basis and express this operator in that basis.

If the new basis vectors are b_1 and b_2 ,
then the operator expressed in this new basis is

$$\begin{pmatrix} 1 & 1 \\ b_1 & b_2 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ b_1 & b_2 \\ 1 & 1 \end{pmatrix}$$

$$BB = BS \quad SS \quad SB,$$

$$\text{and } SB \text{ is } \begin{pmatrix} b_1 & b_2 \\ 1 & 1 \end{pmatrix}.$$

b) Using the standard basis, the vector v is given as $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$. Write v using the basis $b_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$,

$b_2 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$, where these new basis vectors are also given in the standard basis.

$$\begin{pmatrix} 1 & -2 \\ 2 & -2 \end{pmatrix}$$

converts from the new basis
to the standard basis.

so $\begin{pmatrix} 1 & -2 \\ 2 & -2 \end{pmatrix}^{-1}$ converts from the standard
basis to the new basis.

$$\begin{pmatrix} 1 & -2 \\ 2 & -2 \end{pmatrix}^{-1} = \frac{1}{2} \begin{pmatrix} -2 & 2 \\ -2 & 1 \end{pmatrix}$$

so $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ in the new basis is

$$\frac{1}{2} \begin{pmatrix} -2 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 6 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$\text{Indeed, } 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + -1 \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}.$$

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8. a) Compute the determinant of this matrix:

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 1 & 2 \\ 6 & 4 & 10 \end{pmatrix}$$

$$\text{Since } C_3 = C_1 + C_2,$$

$$\det A = 0.$$

b) Give two examples A and B of 3×3 matrices with determinant 20. Then find the determinants of A^2 and B^{-1} .

$$\text{eg. } A = \begin{pmatrix} 20 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$$\det A^2 = 400$$

$$\det B^{-1} = \frac{1}{20}$$

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9. a) For the following matrix, find its eigenvalues and a basis for each of its eigenspaces.

$$A = \begin{pmatrix} 0 & 3 & 0 \\ 3 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 3 & 0 \\ 3 & -\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)[\lambda^2 - 9] = (2-\lambda)(\lambda-3)(\lambda+3)$$

so $\lambda = 2, 3, -3$

~~$\lambda=2$~~ $\rightarrow \text{Nul} \begin{pmatrix} -2 & 3 & 0 \\ 3 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightsquigarrow \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for the 2-eigenspace

$\lambda=3 \rightarrow \text{Nul} \begin{pmatrix} -3 & 3 & 0 \\ 3 & -3 & 0 \\ 0 & 0 & -1 \end{pmatrix} \rightsquigarrow \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$ is a basis for the 3-eigenspace.

$\lambda=-3 \rightsquigarrow \text{Nul} \begin{pmatrix} 3 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} \rightsquigarrow \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\}$ is a basis for the -3-eigenspace.

b) Give an example of two matrices with the same determinant but that have different characteristic polynomials; or explain why this is not possible.

eg. $\begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$ has $\det = 12$ and
char poly = $(6-\lambda)(2-\lambda)$

and $\begin{pmatrix} 12 & 0 \\ 0 & 1 \end{pmatrix}$ has $\det = 12$ and
char poly = $(12-\lambda)(1-\lambda)$

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10. a) Compute an orthonormal basis for the subspace spanned by this set of vectors in $\mathbb{R}^4$:

$$\left\{ \underset{u}{\begin{pmatrix} 1 \\ 0 \\ 2 \\ 2 \end{pmatrix}}, \underset{v}{\begin{pmatrix} 3 \\ 1 \\ 6 \\ 6 \end{pmatrix}} \right\}$$

$$u \cdot v = 3 + 12 + 12 = 27$$

$$u \cdot u = 1^2 + 0^2 + 2^2 + 2^2 = 9$$

$$v - \frac{27}{9}u = v - 3u = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{so } \left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\} \text{ form}$$

an orthogonal basis for the subspace, and

$$\left\{ \begin{pmatrix} 1/3 \\ 0 \\ 2/3 \\ 2/3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

b) What is a basis for the subspace of $\mathbb{R}^4$ that is orthogonal to v ?

$$v = \begin{pmatrix} 4 \\ 12 \\ 0 \\ -2 \end{pmatrix}$$

form an orthonormal basis for the subspace.

$$\text{Nul} \begin{pmatrix} 4 & 12 & 0 & -2 \end{pmatrix} = \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = x \begin{pmatrix} -3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 1/2 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$